INTERNSHIP REPORT

Presented by

Abida Mariem

Internship period: from **01/07/2025** to **31/07/2025** at:

Amilcar Petroleum Operations

Project subject:

Intelligent IT Data Processing and Visualization with Power BI for Ticket System

Internship Supervisor: **Zied Maazoun**

Academic Tutor: **TEK-UP**

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Professional Guidance

First and foremost, I extend my sincere thanks to:

- Mr. Zied Maazoun, IT Infrastructure Applications Engineer and my internship supervisor, for his exceptional guidance, patience, and professional advice throughout my internship. His availability and constructive feedback were instrumental in accomplishing my tasks and personal growth.

Institutional Support

I am grateful to:

- TEK-UP and my professors for providing this internship opportunity and their academic guidance.
- The entire APO staff for their kindness and cooperation, making my internship both enjoyable and enriching.

Personal Appreciation

Finally, I thank my family and friends for their unconditional support throughout my academic and professional journey. Their encouragement and moral strength were essential to the completion of this internship.

Declaration

I hereby declare that this internship report titled "Data Processing and Dashboard Optimization" is my own original work, carried out under the supervision of Mr. Zied Maazoun at Amilcar APO. All the data, analysis, and conclusions presented are based on my personal efforts during the internship period. I confirm that no part of this work has been submitted previously for any other degree or qualification.

Mariem Abida **31/07/2025**

Project Overview:
The main objective of the project was to address and resolve errors occurring in a Power BI dashboard connected to a specific Excel data source. These issues typically arose when the structure of the Excel file was modified—such as adding, deleting, or altering columns or cells—causing the dashboard to fail or display incorrect results.

To solve this, I conducted a comprehensive data treatment process aimed at stabilizing the interaction between Power BI and Excel. After resolving these issues, I designed and implemented a new dashboard template, optimized for consistent performance and more efficient data visualization.

Chapter 1

Company Presentation and Internship Context

1.1 Company Presentation: Amilcar Petroleum Operations (APO)

1.1.1 APO Overview

Amilcar Petroleum Operations Tunisia (APO) is a leading hydrocarbon company specializing in natural gas extraction and production. The company's core activities include:

- Exploration and production
- Storage and distribution
- Natural gas liquefaction
- Power generation
- Gas market development

APO owns and operates one of the world's most extensive gas transportation systems.

The company maintains stringent safety protocols, with all hazardous operations governed by mandatory safety rules across APO subsidiaries.



1.1.2 APO Tunisia Operations

As an APO Group subsidiary, APO Tunisia supplies 60% of the country's natural gas demand. The Tunisian operations began in 1989 with exploration of the Miskar field in the Gulf of Gabès, with first gas production commencing in 1996 at the Hannibal plant.

Current operations span three key assets:

- Amilcar license area
- Miskar concession
- Hasdrubal project

1.2 Environmental Protection

APO Tunisia maintains ISO 14001 certification for environmental management. Site restrictions include:

- No smoking policy
- Prohibited electronic devices:
  - Mobile phones
  - Digital cameras
  - Diagnostic equipment
  - Radio transmitters
  - Calculators



Conclusion

The first chapter was dedicated to presenting APO and its organizational structure. Following this, we provided a description of the internship department and the project theme. The subsequent chapter will focus on analyzing and specifying the technical requirements.

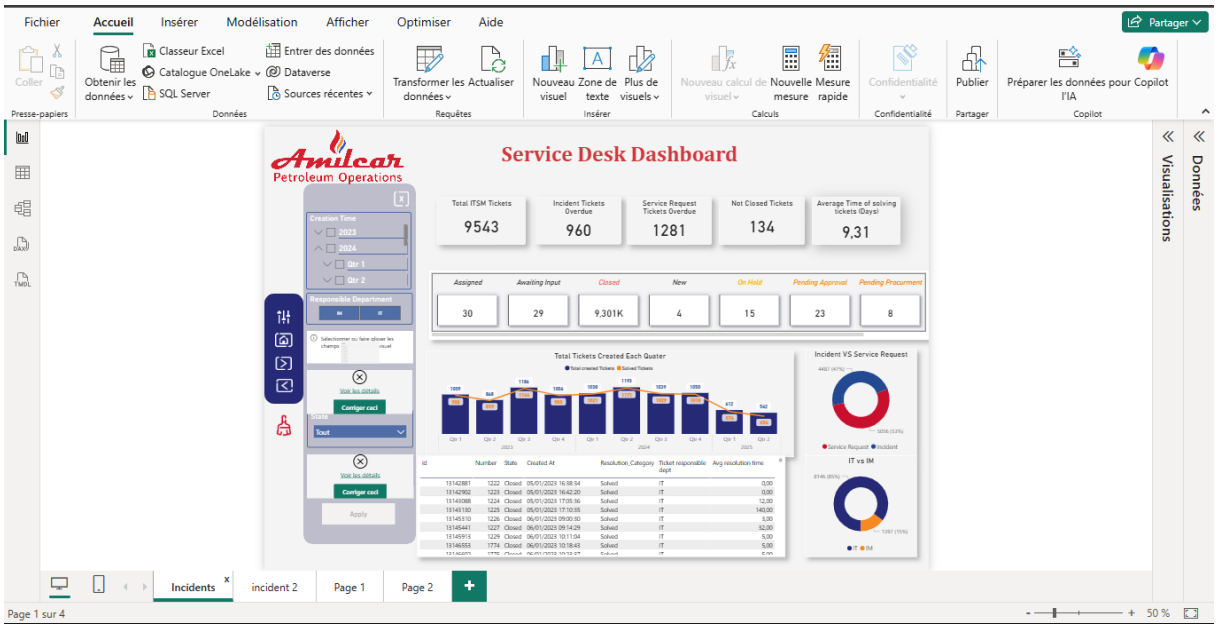
Chapter 2

Work Accomplished

2.1 Work Accomplished

2.1.1 Identified Problem

During the initial phase of the internship, I was assigned to investigate recurring issues within an internal dashboard connected to a specific Excel file. The dashboard was highly sensitive to structural modifications in the source file, such as column deletions, insertions, or value changes. These changes systematically caused errors in the data visualizations and disrupted the functioning of the dashboard.



2.1.2 Situation Analysis

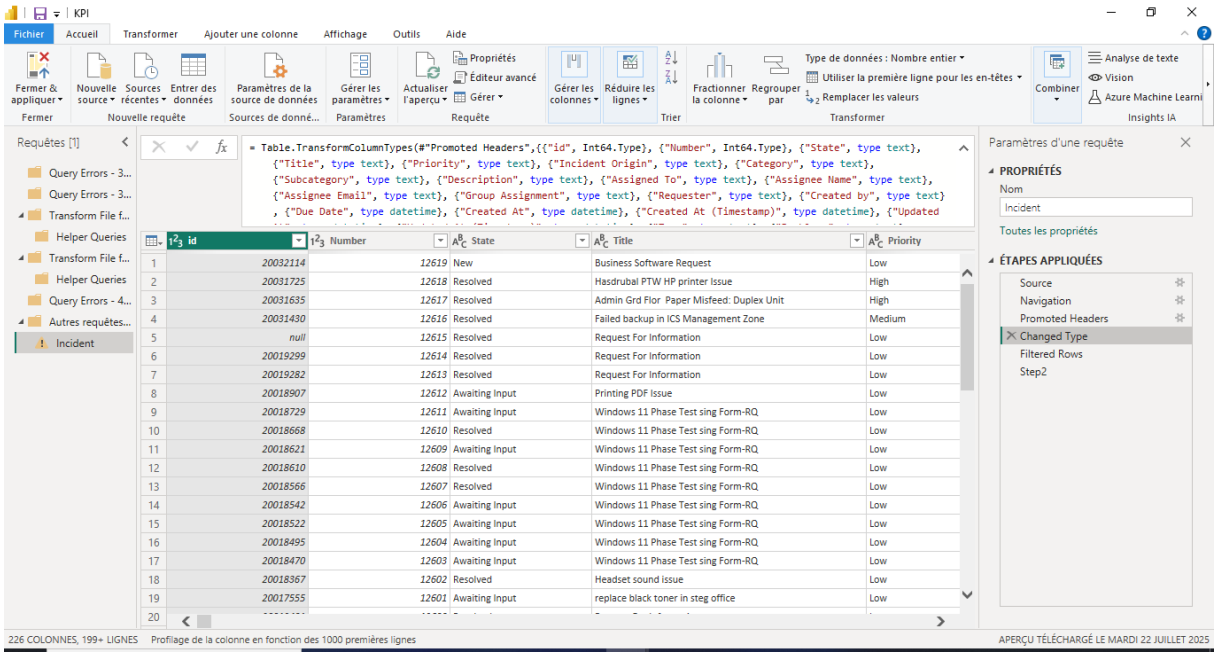
A detailed analysis revealed that the dashboard had strong dependencies on the structure and integrity of the Excel file. Any alteration in the layout (such as renaming, removing, or reordering columns) led to broken references in the dashboard. This indicated a lack of flexibility in the data model and highlighted the need for a more robust and fault-tolerant design.

2.1.3 Data Treatment Methodology

To address these issues, I implemented a comprehensive data treatment process aimed at minimizing structural errors and ensuring consistency between the source data and the dashboard. This involved:

- Establishing validation checks to detect and handle missing or renamed columns.
- Implementing data cleaning routines to handle empty or invalid entries.
- Restructuring the source file to follow a consistent schema.
- Enhancing data security by managing access and limiting direct edits on critical files.

These steps significantly reduced the frequency of errors and improved the reliability of the dashboard.

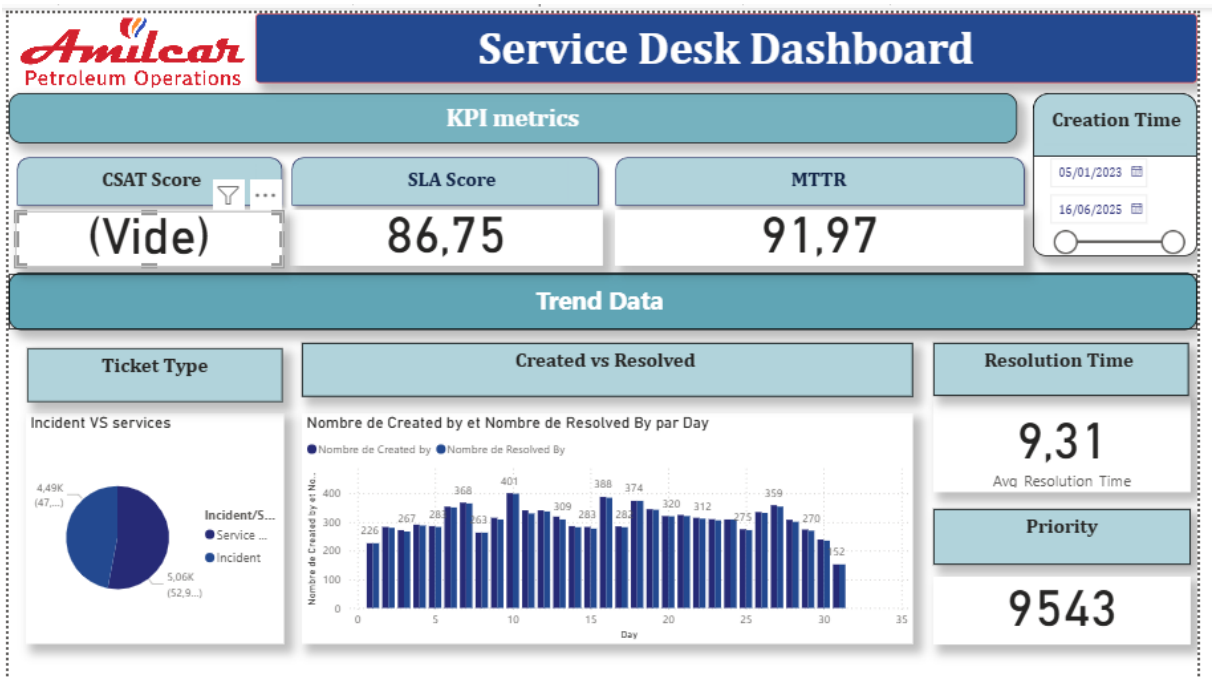


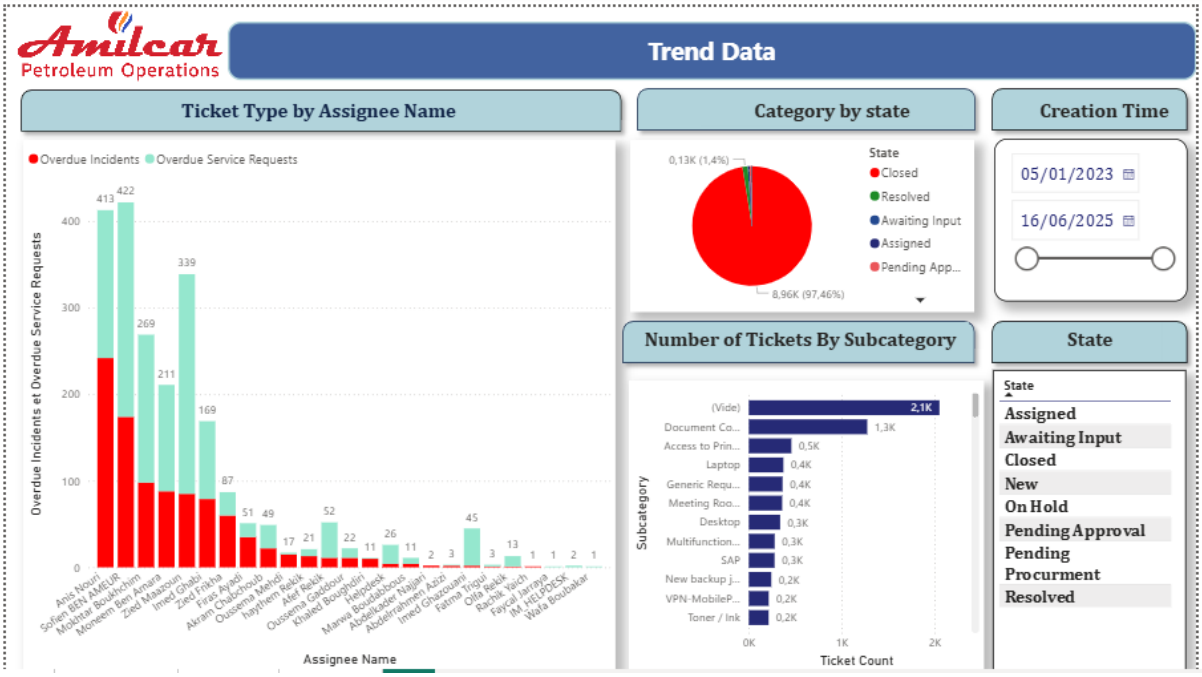
2.1.4 Creation of the New Dashboard

After stabilizing the existing dashboard, I initiated the development of a new dashboard based on a redesigned data model. The goal was to create a more sustainable and user-friendly solution. Key improvements included:

- A new template with dynamic column mapping to reduce hardcoded dependencies.
- Clear separation between raw data, treated data, and final visualizations.
- Integration of meaningful KPIs aligned with business objectives.
- Use of interactive visuals and filters to enhance user experience.

This new dashboard offers greater flexibility, improved performance, and reduced risk of failure when the data structure evolves.





When the user clicks on the **Trend Data** button in the first dashboard, they are automatically redirected to the second dashboard displaying detailed charts related to trends.

- **CSAT Score (Customer Satisfaction Score)**: Measures customer satisfaction with the support services provided.
- **SLA Score (Service Level Agreement Score)**: Indicates the team's adherence to agreed service level targets.
- **MTTR (Mean Time to Resolve)**: Represents the average time taken to resolve tickets, reflecting efficiency in issue resolution.

Chapter 3

Internship Summary

3.1 Technical and Professional Skills Acquired

Throughout this internship, I enhanced both my technical proficiency and my professional mindset. I became proficient in **Microsoft Excel**, focusing on structured data management, advanced formulas, and cleaning techniques. I also developed strong skills in **Power BI**, particularly in dashboard design, data transformation with Power Query, KPI visualization, and report automation.

One of the key technical competencies acquired was mastering **Power Query**, which allowed me to preprocess and clean data efficiently before visualization. In addition, I learned how to apply **data modeling techniques** and create relationships between tables to produce dynamic and interactive dashboards.

On the professional side, I developed greater autonomy in managing tasks, improved my problem-solving approach, and strengthened my collaboration and communication within the team under the guidance of Mr. Zied Maazoun.

3.1.1 Challenges Encountered and Solutions Provided

During the internship, I faced multiple challenges, particularly related to the **insertion of new Excel files** into the Power BI system. When a new file was introduced, it often contained structural changes such as:

- Renamed columns,
- Removed or additional columns,
- Inconsistent data types.

These issues caused significant errors in the dashboard.

To address these problems, I worked extensively in **Power Query** to adapt the data preprocessing steps. This involved:

- Manually adjusting the applied steps, especially the `Changed Type` step, which often broke due to unexpected columns.
- Eliminating unnecessary columns to simplify the dataset and reduce noise.
- Reordering and renaming columns to match the expected schema.
- Standardizing data types to avoid type mismatch errors during transformations.

Thanks to these actions, I was able to make the data pipeline more stable and the dashboard less error-prone.

When creating the **new dashboard**, I encountered additional challenges related to **data classification and presentation**. One major difficulty was how to group and categorize data meaningfully to provide clear and actionable insights. This required:

- Redesigning the data model to separate raw data from cleaned data.
- Identifying key business indicators (KPIs) relevant to the stakeholders.
- Defining logical dimensions and hierarchies (e.g., category, subcategory, time period).
- Building visuals that are both informative and user-friendly, such as bar charts, KPIs, and slicers.

Through multiple iterations and validations, I succeeded in creating a dashboard that not only corrected the problems of the previous version but also brought added value to the company in terms of usability, flexibility, and clarity.

3.1.2 Limits and Perspectives for Improvement

While the final dashboard represented a clear improvement, certain limitations remain. For example:

- The system still relies on manual updates of Excel files, which can lead to human error.
- There is no automated validation process when new files are inserted.
- The scalability of the current setup is limited if the volume of data increases significantly.

Future improvements could include:

- Replacing the Excel source with a **centralized database** or **cloud-based data connector**.
- Implementing **automated validation rules** and data integrity checks.
- Adding **user-specific views and permissions** to customize dashboards based on roles.
- Integrating **advanced analytics features**, such as trends, forecasts, and alerts, to support strategic decision-making.

3.2 Results and Impact

3.2.1 Operational Improvements

The implementation of the new dashboard brought significant improvements to daily operations:

- **Error Reduction and Stability:** Transitioned from a fragile dashboard prone to errors after Excel file modifications to a more robust system capable of handling structural changes smoothly.
- **Improved Data Quality:** Automated data cleaning and validation steps reduced inconsistencies and manual correction efforts.
- **Faster Issue Detection:** Enabled quicker identification and resolution of data mismatches or missing columns, minimizing downtime and improving reliability.

3.2.2 Strategic Benefits for APO

The solution delivered important benefits at the organizational level:

- **Enhanced Data Governance:** Standardized data formats and cleaning procedures help maintain data integrity across teams.
- **Resource Optimization:** Streamlined reporting and visualization processes reduce time spent on manual data manipulation.
- **Scalable Framework:** Established a reusable data treatment and dashboard template adaptable to future projects or datasets.
- **Training and Knowledge Sharing:** The project served as a practical example and learning resource for colleagues and new staff.

3.3 Process Documentation

The dashboard development followed a structured workflow:

1. **Data Extraction:**
    - Consolidated multiple Excel files and sheets into a standardized master dataset.
    - Archived original files to ensure traceability and enable rollback if necessary.
2. **Data Cleaning:**
    - Detected and removed invalid or irrelevant data entries that could skew analysis.
    - Standardized column names and data types to ensure uniformity and compatibility.
3. **Power BI Modeling:**
    - Created key measures and calculated columns using DAX formulas for essential KPIs.
    - Built data relationships and hierarchies to support dynamic filtering and drill-down.
4. **Validation:**
    - Conducted thorough testing with input from supervisors and domain experts to verify accuracy.
    - Compared dashboard results against manual calculations and source data to ensure correctness.

3.4 Final Remarks

This project has demonstrated how digital transformation and automation in data handling can greatly enhance operational efficiency and reporting reliability at APO. The implemented dashboard not only resolved immediate issues related to data errors but also provided a solid foundation for future data-driven decision making.

The skills and knowledge gained through this internship contribute significantly to my professional development as an engineer specializing in data processing and business intelligence within the energy sector.

Chapter 4

Field Visits and Observations

During my internship, I had the opportunity to visit several operational sites, which enriched my understanding of the company's infrastructure and processes. These visits allowed me to observe the intersection between IT systems and industrial operations, reinforcing the importance of software solutions in optimizing and securing critical workflows. Below are the key takeaways from my three visits:

4.1 Visit to the Data Center: Network and Security Infrastructure

My first visit was to the company's data center, where I observed the hardware responsible for database management and network security, including firewalls and servers. This was my first time seeing such systems in a real-world production environment, and it deepened my understanding of:

- **Network Management:** How data flows between systems and the role of routing/switching in maintaining efficiency.
- **Data Security:** The physical and software mechanisms (firewalls, intrusion detection systems) that protect sensitive company data.
- **IT-Industrial Synergy:** The reliance of operational sites on robust IT infrastructure for real-time monitoring and data integrity.

As a software developer, this visit highlighted the importance of writing secure, efficient code that interacts seamlessly with hardware—especially in environments where data reliability is critical (e.g., industrial control systems).

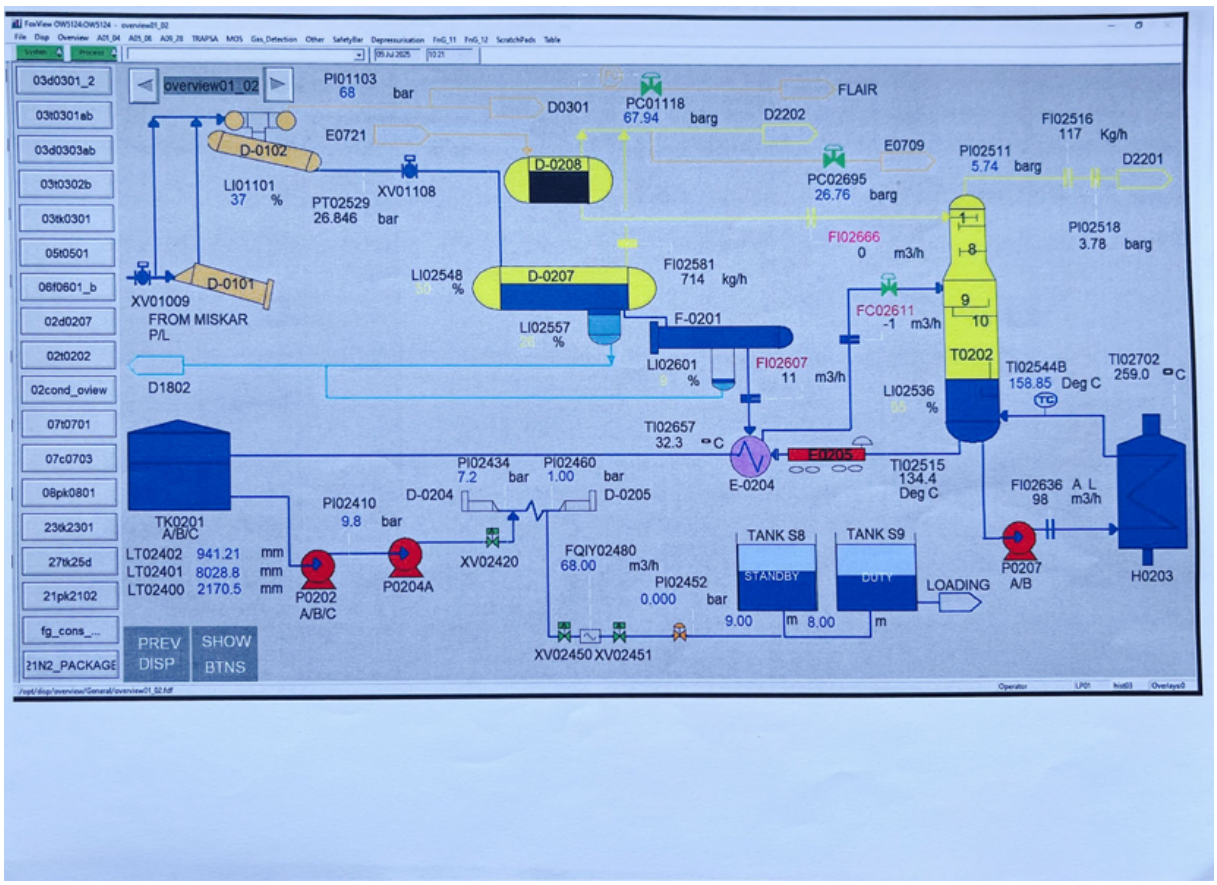
4.2 Visit to the Hasdrubal Control Room: Process Automation

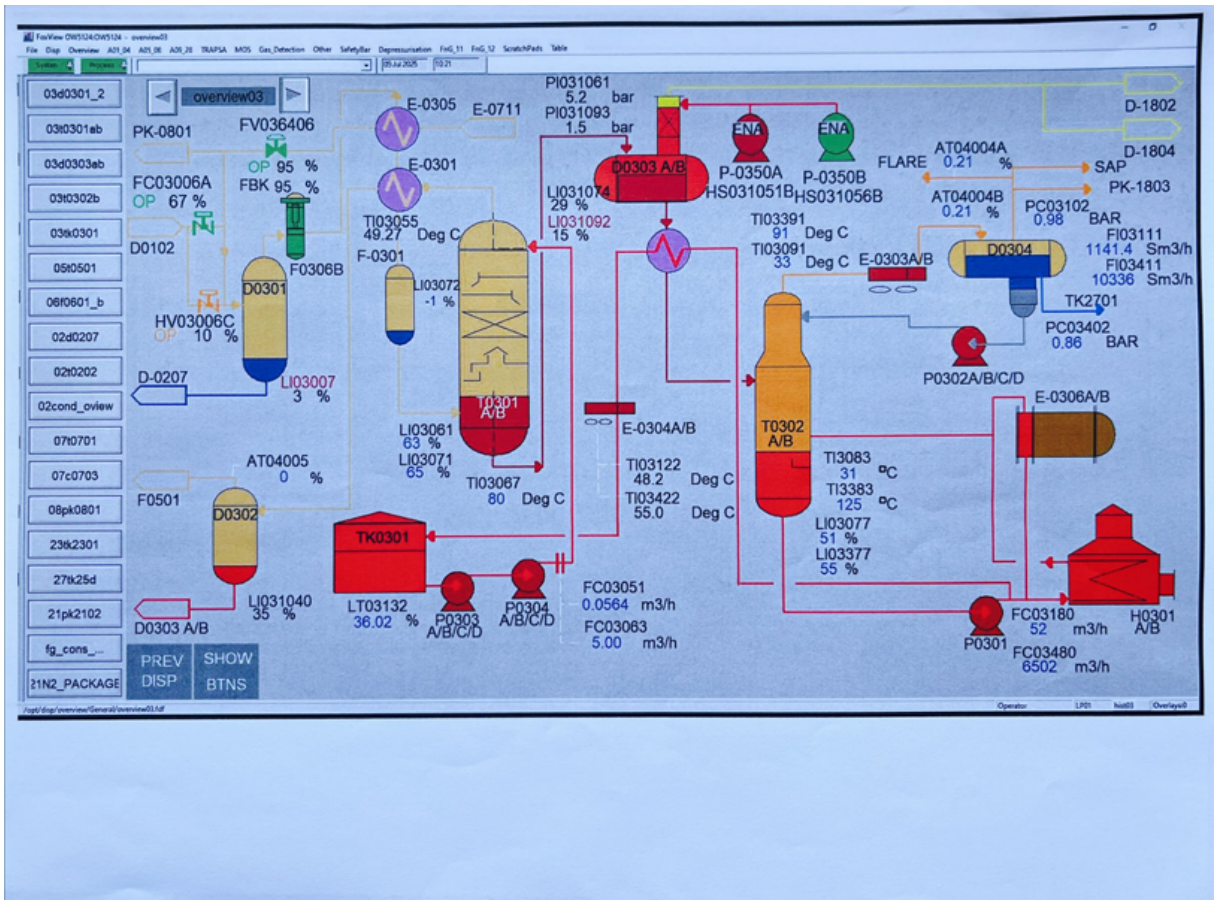
In my second visit, I observed the Hasdrubal control room, where raw gas (from Miskar) undergoes purification. Key steps included:

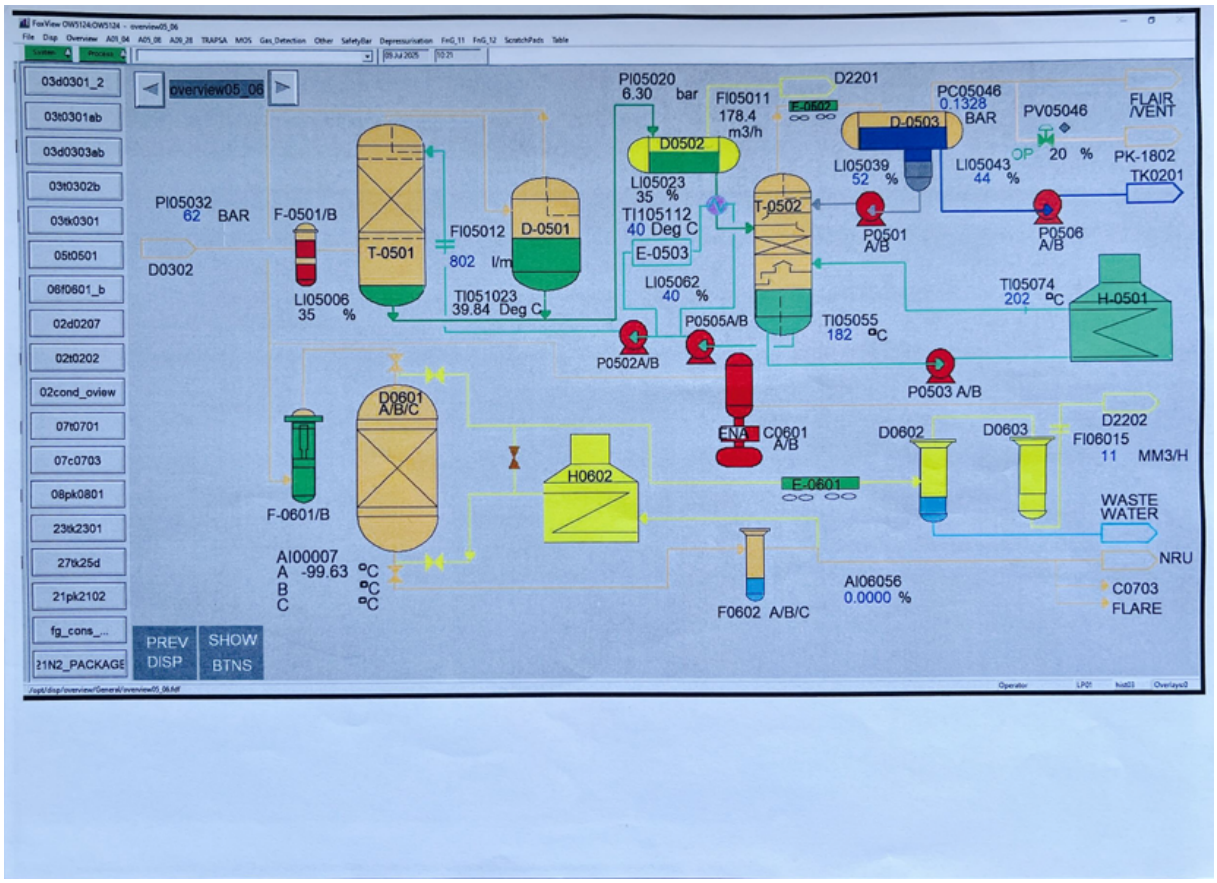
- **Sweetening Process:** Removal of toxic H_2S and CO_2 using amine solutions to prevent corrosion.
- **Nitrogen Rejection Unit (NRU):** Separation via distillation, leveraging nitrogen's volatility.

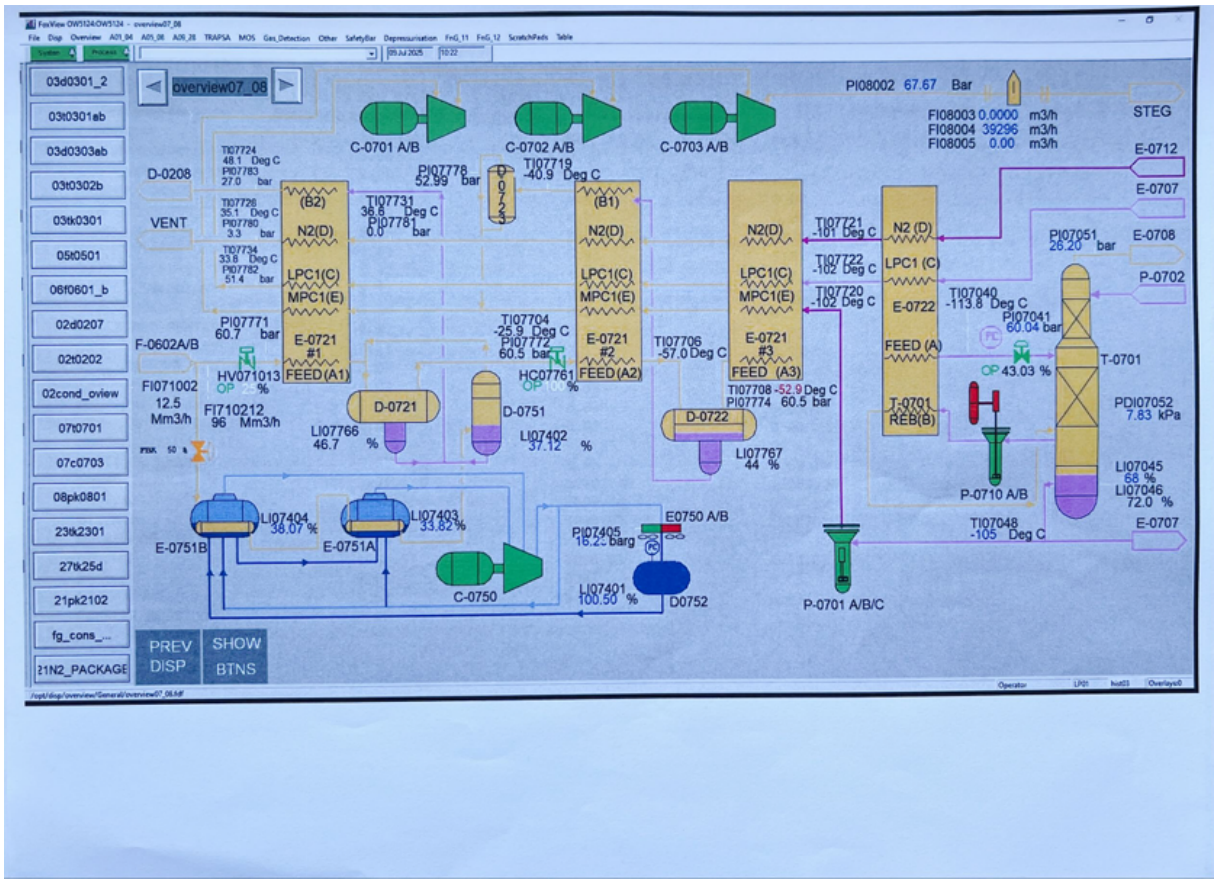
The entire process was monitored and controlled through Fox View, a supervisory software displaying real-time metrics. This visit was particularly insightful because:

- It demonstrated how software interfaces (like Fox View) bridge the gap between engineers and physical processes.
- It underscored the role of automation in ensuring precision (e.g., maintaining $H_2S < 0.6$ ppm).
- It revealed opportunities for software optimization (e.g., predictive maintenance algorithms, data visualization enhancements).




overview03
PK-0801
FV036406
OP 95 %
FBK 95 %
FC03006A
OP 67 %
D0102
F0306B
D0301
HV03006C
OP 10 %
LI03007
3 %
D-0207
E-0305
E-0711
E-0301
TI03055
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F-0301
LI03072
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T0301 A/B
LI03061
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LI03071
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PI031093
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D0303 A/B
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D0304
TK2701
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P0302A/B/C/D
E-0306A/B
E-0304A/B
TI03122
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TI03422
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T0302 A/B
TI3083
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TI3383
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T-0502
LI05039
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LI05043
44 %
OP 20 %
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TK0201
P0501 A/B
P0506 A/B
PI05032
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F-0501/B
T-0501
FI05012
802 l/m
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202 °C
H-0501
P0503 A/B
D0601 A/B/C
C0601 A/B
D0602
D0603
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H0602
E-0601
WASTE WATER
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08pk0801
23tk2301
27tk25d
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fg_cons_...
21N2_PACKAGE
PREV DISP
SHOW BTNS



4.3 On-Site Visit to Hasdrubal: Practical Implementation

My final visit was an on-site tour of Hasdrubal's operational units. Unlike the control room's digital perspective, this visit emphasized:

- **Safety Protocols:** Wearing PPE and gas detectors, adhering to strict operational guidelines.
- **Human-Machine Interaction:** How field technicians relay manual data to control systems, highlighting the need for user-friendly software to reduce errors.
- **Real-World Constraints:** Environmental factors (e.g., temperature, pressure) that software must account for in industrial settings.

4.4 Connections to Software Development

These visits reinforced how integral software is to industrial operations:

1. **Security:** My work in IT aligns with safeguarding data flows between control systems and databases.
2. **Automation:** Developing tools to streamline data logging or predictive analytics could enhance process efficiency.
3. **User Interfaces:** Observing Fox View inspired ideas for more intuitive dashboards or alert systems for operators.

Conclusion:
Stepping outside my typical office environment gave me a holistic view of the company's operations. It underscored that effective software solutions must account for hardware limitations, safety protocols, and real-time data needs—a perspective I will carry forward in my career as a developer.

Chapter 5

General Conclusion

The internship at Amilcar Petroleum Operations (APO) focused on developing a resilient Power BI dashboard to manage and visualize data sourced from Excel files, addressing challenges caused by file structure modifications and data inconsistencies.

Key outcomes include:

- **Enhanced Operational Efficiency:** Reduced errors and improved responsiveness through automation and data cleaning.
- **Strategic Benefits:** Improved data governance and established a scalable and adaptable dashboard framework.
- **Knowledge Transfer:** The project provided valuable training opportunities and highlighted the importance of data-driven decision making.

Overall, this experience showcases how leveraging digital tools in industrial contexts can deliver tangible improvements and foster professional growth by solving complex data challenges.